

Vulnerability Assessment Report

Future Interns - Cyber Security Internship Task 1

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Objective

- Identify basic security vulnerabilities in the target website
- Analyze open ports and security misconfigurations
- Perform passive vulnerability assessment using OWASP ZAP
- Understand common web application security risks
- Document findings and provide remediation recommendations

Tools Used

- OWASP ZAP – Passive vulnerability scanning
- Nmap – Network and port scanning
- Browser DevTools – Website inspection and analysis
- Canva – Report design and documentation

Target Website Information

Website Tested:

<http://zero.webappsecurity.com>

Purpose:

This website is a publicly available demo application used for security testing and vulnerability assessment practice in a safe and legal environment.

Assessment Type:

Passive Vulnerability Assessment

Findings Summary

Vulnerability	Risk Level	Status
CSP Header Not Set	Medium	Detected
Missing Anti-clickjacking Header	Medium	Detected
Vulnerable JS Library	Low	Detected
Cross-Domain Misconfiguration	Medium	Detected
X-Content-Type-Options Header Missing	Low	Detected

Vulnerability Details

Vulnerability:

Content Security Policy (CSP) Header Not Set

Risk Level:

Medium

Description:

The website does not implement a Content Security Policy (CSP) header. This may increase the risk of Cross-Site Scripting (XSS) and malicious content injection attacks.

Recommendation:

Implement proper Content Security Policy headers to restrict unauthorized scripts and improve browser security protections.

Nmap Scan Results

The Nmap scan identified several open ports and active services running on the target website.

Detected Open Ports:

- Port 53 – Domain Service
- Port 80 – HTTP
- Port 443 – HTTPS
- Port 8080 – HTTP Proxy

These ports indicate active web and network services accessible from the internet.

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\gmadh>nmap testphp.vulnweb.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-22 22:37 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.95 seconds

C:\Users\gmadh>nmap zero.webappsecurity.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-22 22:57 +0530
Nmap scan report for zero.webappsecurity.com (54.82.22.214)
Host is up (0.37s latency).
rDNS record for 54.82.22.214: ec2-54-82-22-214.compute-1.amazonaws.com
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
8080/tcp   open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 36.89 seconds

C:\Users\gmadh>
```

Security Assessment Evidence

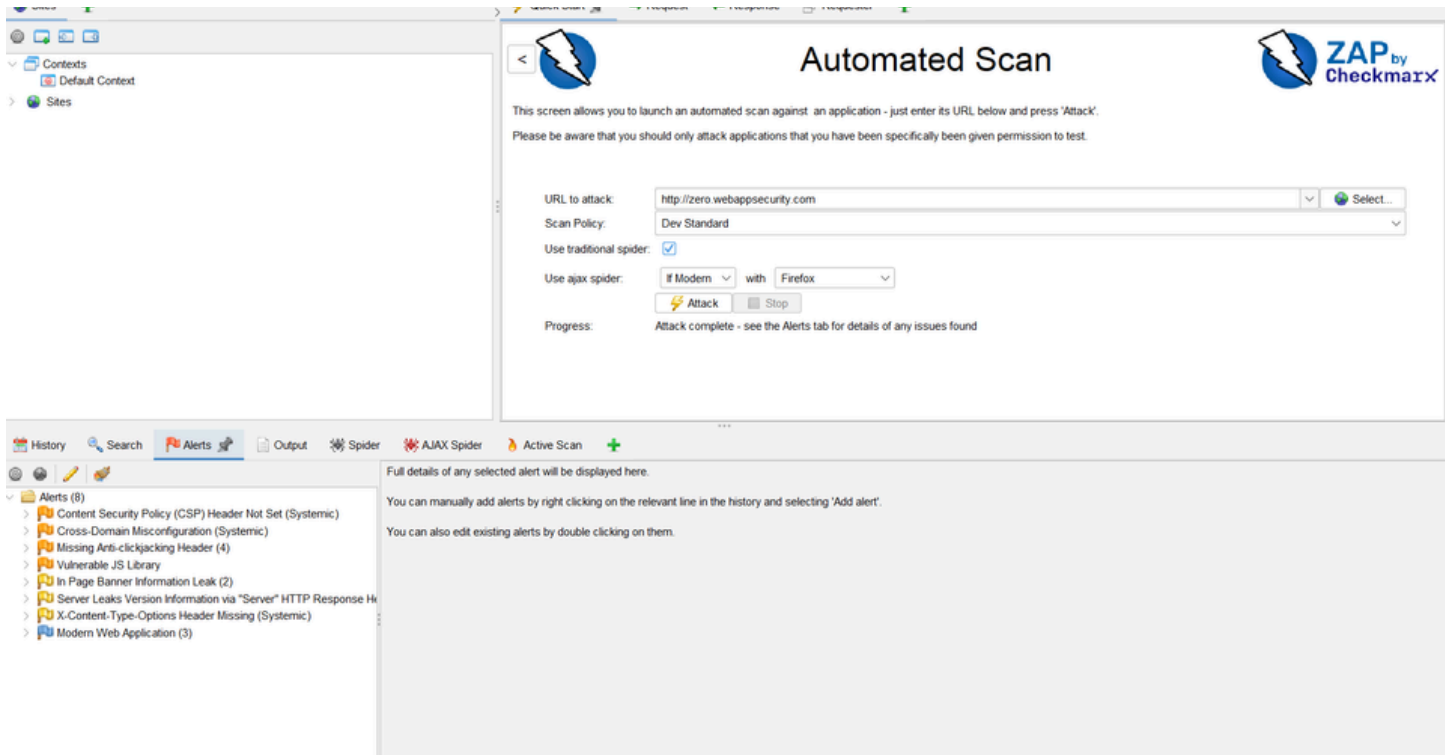


Figure 1: Successful OWASP ZAP Scan Completion

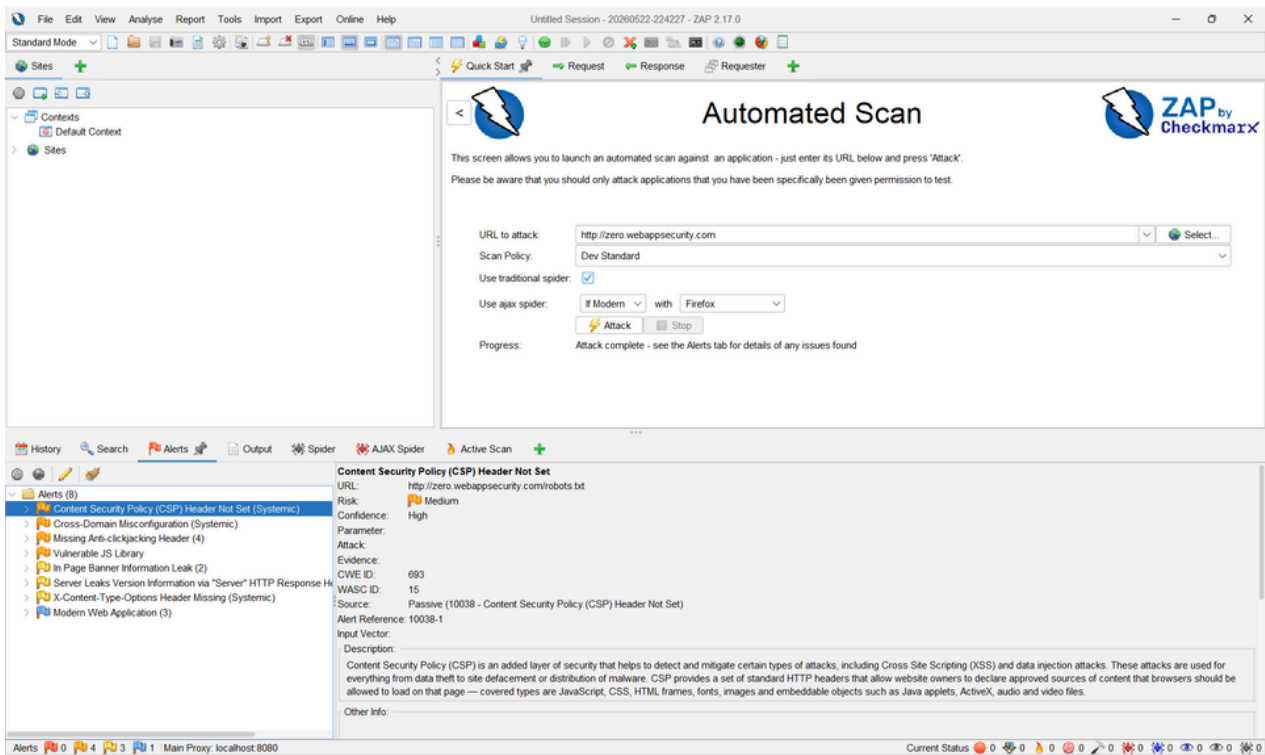


Figure 2: CSP Header Vulnerability

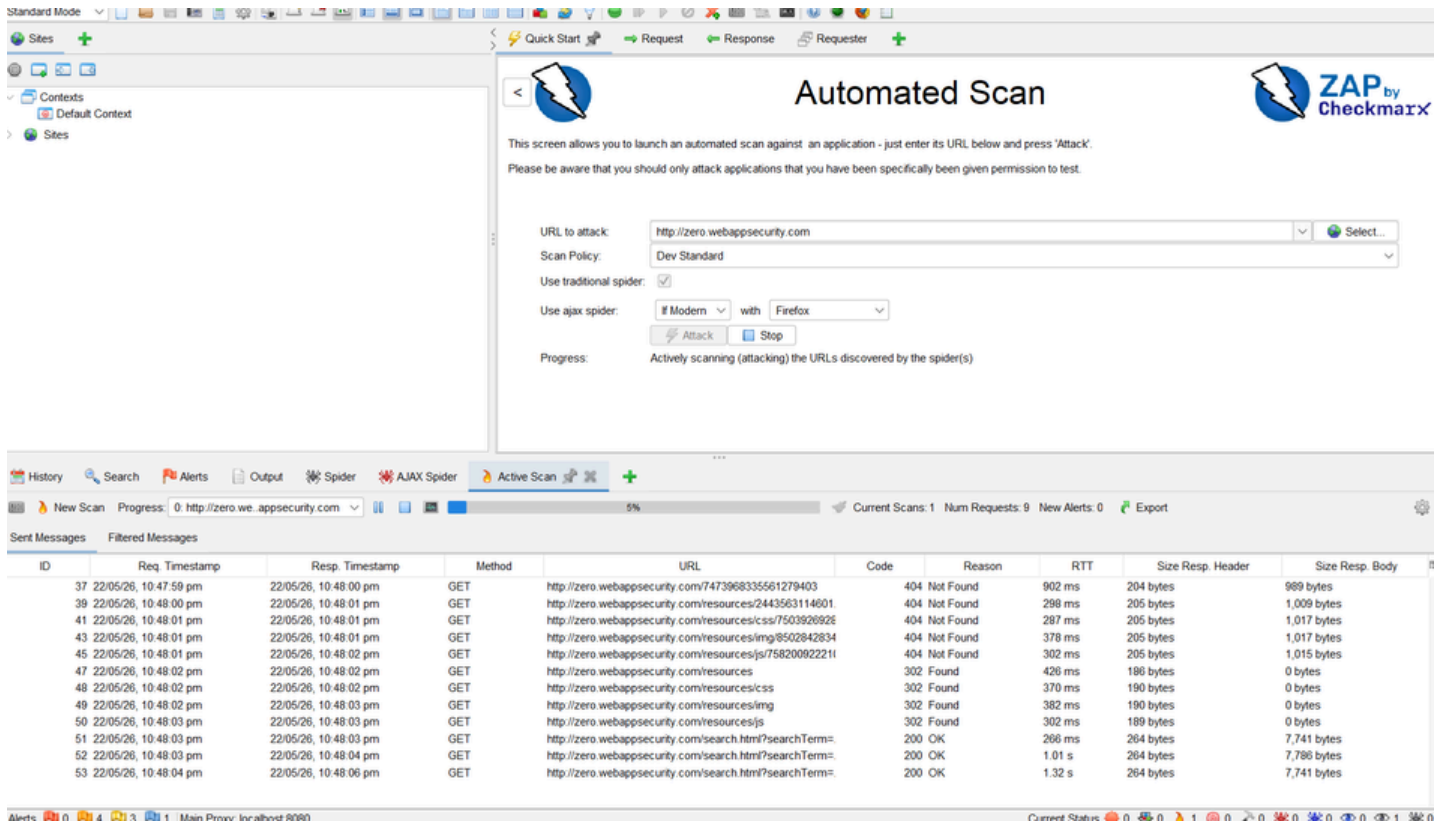


Figure 3: OWASP ZAP Active Scan Process

Security Recommendations

- Implement proper Content Security Policy (CSP) headers.
- Configure X-Frame-Options to prevent clickjacking attacks.
- Update vulnerable JavaScript libraries regularly.
- Configure secure HTTP response headers.
- Restrict unnecessary open ports and services.
- Perform regular vulnerability assessments and security audits.
- Keep web applications and server software updated.

Conclusion

This vulnerability assessment was successfully performed using OWASP ZAP and Nmap on the target web application. Several security weaknesses and misconfigurations were identified, including the absence of security headers and exposed services. The assessment helped in understanding common web application vulnerabilities and the importance of implementing secure configurations. Proper remediation and regular security testing can significantly improve the overall security posture of web applications.

References

- OWASP ZAP Official Documentation
- Nmap Official Documentation
- OWASP Top 10 Web Security Risks
- VulnWeb / Zero WebAppSecurity Test Environment
- Cyber Security Internship Learning Resources